5.1

Essentially the same as the provided solution but the sample solution provides more comments on what is being done at least for the part where we calculate the expectation. Additionally, we calculated the covariance through suitable transformations while the sample solution uses the definition but it is mathematically equivalent.

Take derivative and set to zero:

$$-2X^T(y - X\beta) + 2\tau\beta = X^T X\beta - X^T y + \tau\beta = 0$$

$$\Leftrightarrow (X^T X + \tau I_D)\beta = X^T y$$

We get: $\hat{\beta}_\tau = (X^T X + \tau I_D)^{-1} X^T y$

$$\overset{S_\tau}{=} S_\tau^{-1} X^T y$$

$$\overset{y}{=} S_\tau^{-1} X^T (X\beta^* + \varepsilon)$$

$$= S_\tau^{-1}(X^T X\beta^* + X^T \varepsilon)$$

$$\overset{S}{=} S_\tau^{-1}(S\beta^* + X^T \varepsilon)$$

$$= S_\tau^{-1} S\beta^* + S_\tau^{-1} X^T \varepsilon$$

$\hat{\beta}_\tau$ is random only through ε : $\mathbb{E}[\varepsilon] = 0 \Rightarrow \mathbb{E}[\overbrace{S_\tau^{-1} X^T}^{\text{constant}}\varepsilon] = S_\tau^{-1} X^T \mathbb{E}[\varepsilon] = 0$

$$\Rightarrow \mathbb{E}[\hat{\beta}_\tau] = \mathbb{E}[S_\tau^{-1} S\beta^*] + \mathbb{E}[S_\tau^{-1} X^T \varepsilon] = S_\tau^{-1} S\beta^*$$

$$\mathrm{Cov}[\hat{\beta}_\tau] = \mathrm{Cov}[\overbrace{S_\tau^{-1} X^T}^{\text{constant}}\varepsilon]$$

$$= S_\tau^{-1} X^T \mathrm{Cov}[\varepsilon] (S_\tau^{-1} X^T)^T$$

$$= S_\tau^{-1} X^T \mathrm{Cov}[\varepsilon] X S_\tau^{-1}$$

$$= S_\tau^{-1} X^T (\sigma^2 I_D) X S_\tau^{-1}$$

$$= S_\tau^{-1} (\sigma^2 X^T I_D X) S_\tau^{-1}$$

$$= S_\tau^{-1} S S_\tau^{-1} \sigma^2$$